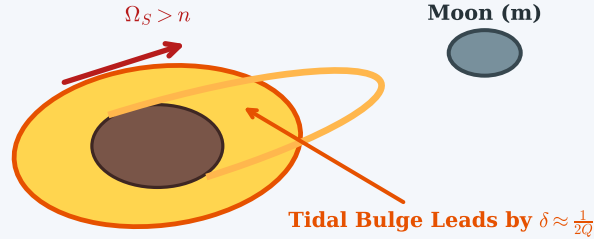
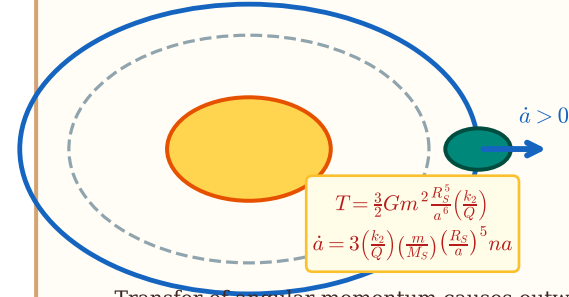


1. Saturn Rotation & Tidal Bulge Lead



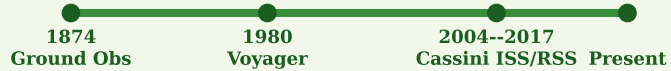
Saturn rotates in 10.65 h ($\Omega_S = 1.64 \times 10^{-4}$ rad/s), faster than inner moons ($P_{\text{orb}} \geq 0.94$ d). Bulge exerts forward torque.

2. Tidal Torque & Orbital Expansion



Transfer of angular momentum causes outward migration and secular orbital deceleration ($\dot{n} < 0$).

3. Astrometric Measurement Baseline

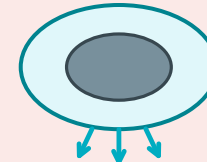


Orbital Longitude Drift: $\Delta\lambda(t) = \frac{1}{2} \dot{n} t^2$

$\dot{n}/n \approx -1.5 \times 10^{-17} \text{ s}^{-1} \Rightarrow Q \approx 1800$

Over 130 years of Earth astrometry + Cassini ephemerides reveal tidal dissipation $10 \times$ stronger than classical limit.

4. Enceladus Resonance & Plume Heat



Tiger Stripe Plumes (15.8 GW)

2:1 Mean-Motion Resonance w/ Dione maintains $e \approx 0.0047$

Steady-state balance: tidal orbital expansion feeds continuous viscoelastic heating of Enceladus ocean.